ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE C

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET second paper (Fritzing practical, 80 marks): a circuit built from a written description, two circuit diagrams built on a breadboard, and a circuit diagram drawn from a breadboard layout.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of TWO questions. Answer ALL the questions.
2. This is a practical paper. Use the Fritzing software for all circuits and breadboard layouts. If you do not have a computer, draw the circuits and breadboard layouts neatly by hand.
3. Put your name or examination number on every printout or page, and place the printouts in your examination folder.
4. Show the positive and negative rails clearly and make sure every component has the correct polarity.
5. Write neatly and legibly.

QUESTION 1

1.1 Use Fritzing to draw a circuit with the following specifications: (20)

• Two LEDs, each with its own 330 Ω series resistor, are connected in parallel with each other.

• A push-to-make button is connected in series with this parallel group.

• A 1N4007 diode is connected in series with the supply, so that the LEDs are protected if the battery is connected the wrong way round.

• The circuit is supplied by a 9 V battery.

1.2 Build the circuit diagram below on a breadboard using Fritzing. When the button is pressed and released, the LED stays on for a few seconds while the capacitor discharges. (20)

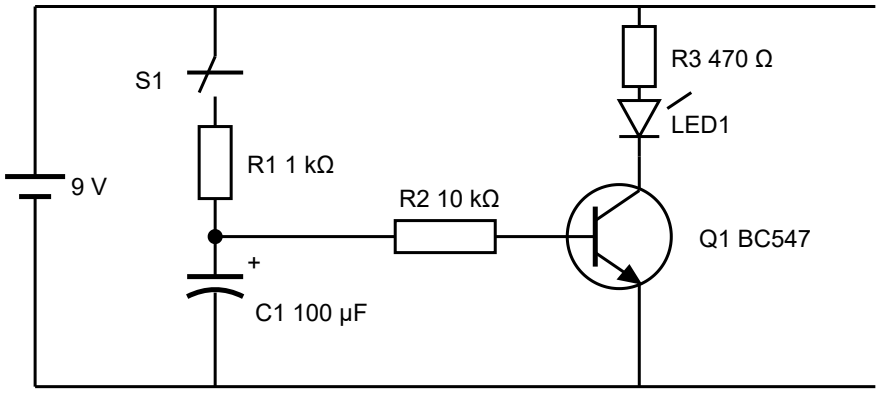


FIGURE 1: LED delay-off timer

[40]

QUESTION 2

2.1 Build the circuit diagram below on a breadboard using Fritzing. Pressing S1 switches on the transistor, which energises the relay; the relay contacts switch on LED2 from the same battery. (20)

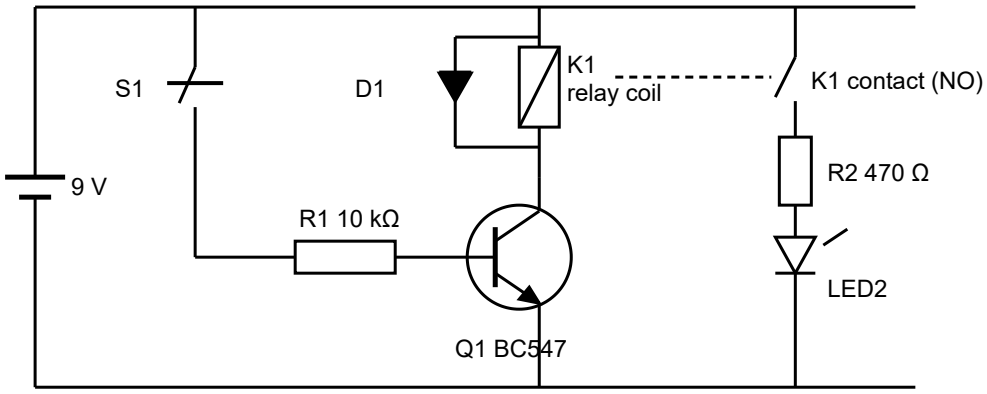


FIGURE 2: Transistor-driven relay (D1 is connected across the relay coil, cathode to +9 V)

2.2 A learner built the circuit below on a breadboard. The table describes where every component is placed. Use Fritzing (or draw by hand) to draw the CIRCUIT DIAGRAM (schematic) of this breadboard layout, using the correct symbols and labels. (20)

Component	Where it is connected on the breadboard
9 V battery	+ lead to the top (+) rail; − lead to the bottom (−) rail
Slide switch S1	one pin in the + rail, other pin in row 5
10 kΩ potentiometer VR1	one outer pin in row 5, wiper in row 9 (other outer pin not connected)
220 Ω resistor R1	from row 9 to row 14
Red LED	anode (long leg) in row 14, cathode in row 18
Jumper wire	from row 18 to the − rail
Active buzzer	+ pin in row 9, − pin in the − rail

[40]

TOTAL: 80

END OF QUESTION PAPER